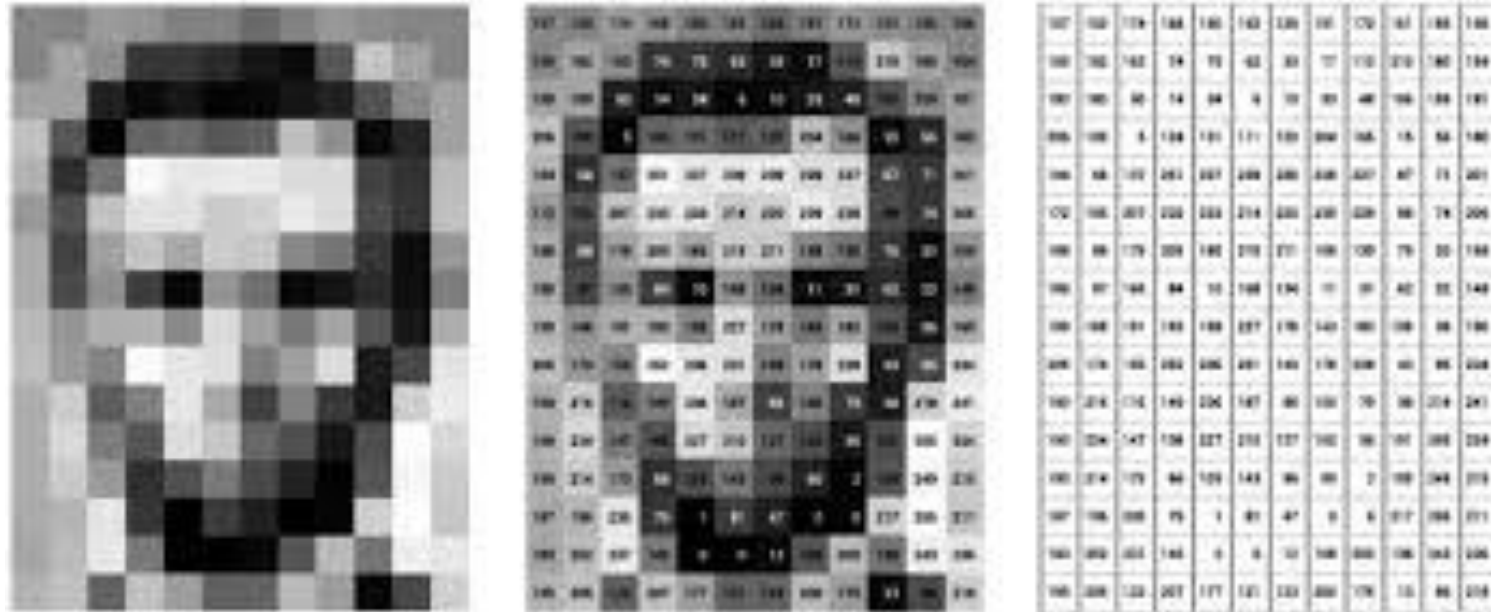


Convolutional Neural Network

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Processing Images with Neural Networks

- We need special neural networks for processing images and grid-like structures.



Convolutional Neural Networks

- Convolutional Neural Networks (CNNs) are a class of deep learning models designed to process **grid-structured data**, especially **images**.
- They **automatically learn spatial features** (edges, textures, shapes) directly from raw data, reducing the need for manual feature engineering.
- CNNs use **convolutional filters** that focus on local regions, exploiting spatial locality.
- CNNs are widely used in image classification, object detection, segmentation, and medical imaging.

Convolution operation

Source layer

5	2	6	8	2	0	1	2
4	3	4	5	1	9	6	3
3	9	2	4	7	7	6	9
1	3	4	6	8	2	2	1
8	4	6	2	3	1	8	8
5	8	9	0	1	0	2	3
9	2	6	6	3	6	2	1
9	8	8	2	6	3	4	5

Convolutional
kernel

-1	0	1
2	1	2
1	-2	0

Destination layer

	5						

$$\begin{aligned} &(-1 \times 5) + (0 \times 2) + (1 \times 6) + \\ &(2 \times 4) + (1 \times 3) + (2 \times 4) + \\ &(1 \times 3) + (-2 \times 9) + (0 \times 2) = 5 \end{aligned}$$

Convolution operation (contd.)

- A small **filter (kernel)** slides over the image to look at local regions.
- At each position, the filter multiplies and sums pixel values to produce one output value.
- The result is a **feature map** that **highlights patterns** like edges or textures.
- The same filter is reused across the whole image, enabling consistent pattern detection.

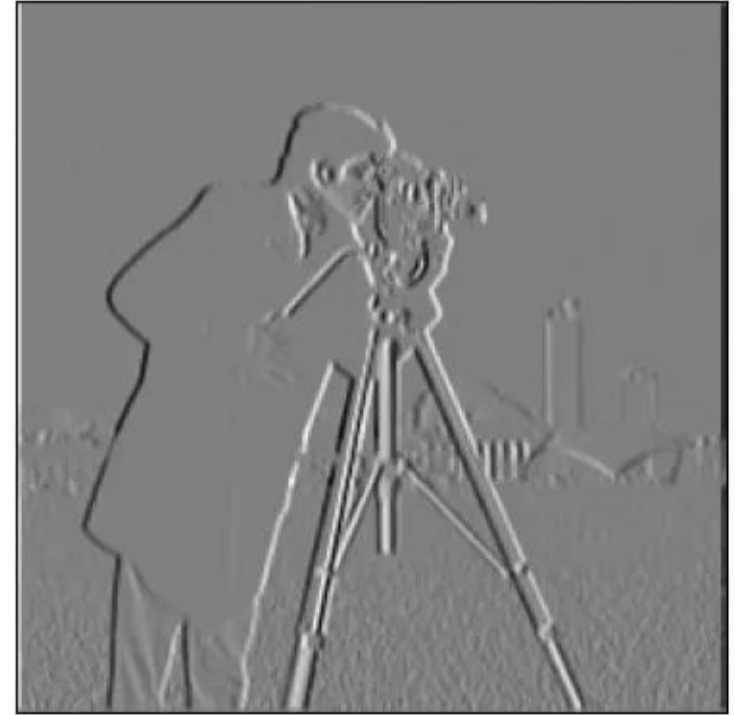
Example



*

1	0	-1
1	0	-1
1	0	-1

=



Convolution with this particular kernel detects **vertical edges**

In CNN, kernels are “learned”

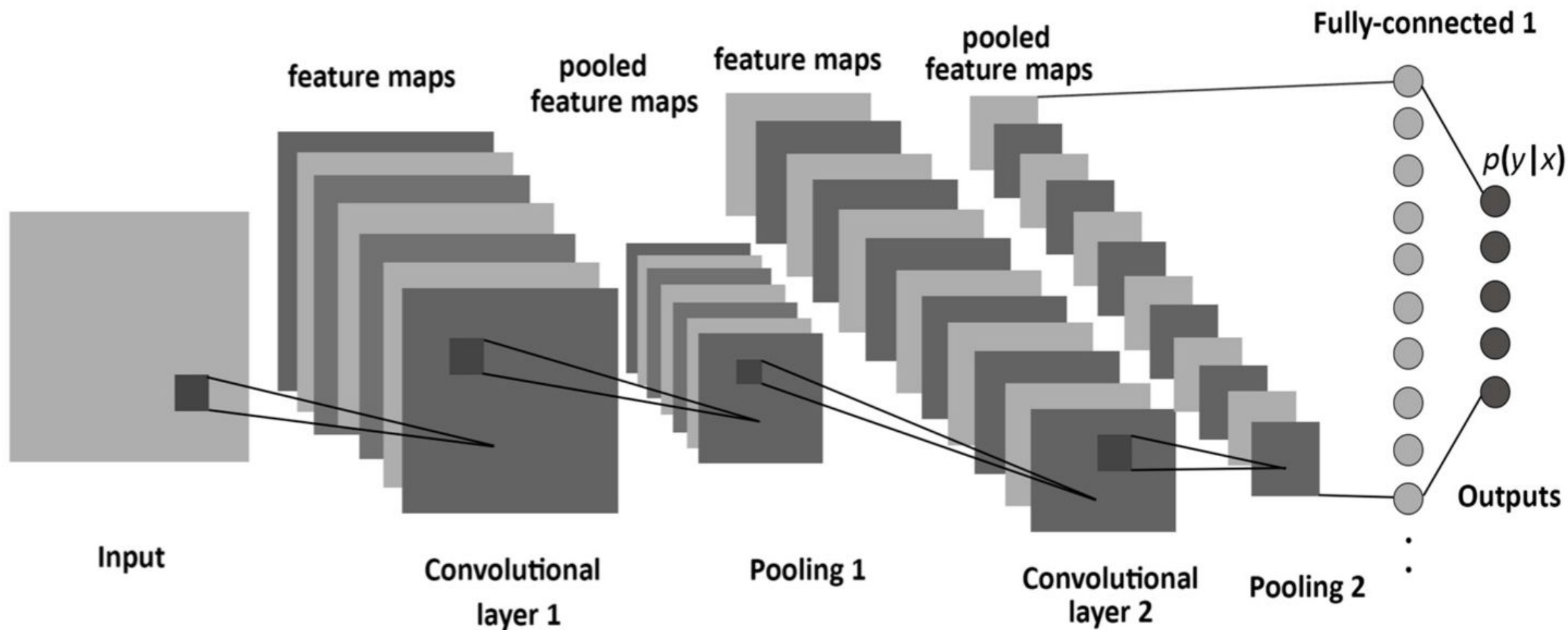
In a Convolutional Neural Network:

- Kernels **start with random values**
- During training, the **network learns the best kernel values**
- Learning happens via **backpropagation + gradient descent**
- The goal is to minimize task loss (e.g., classification error)

Convolutional network

- The key is to remember that convolutions give us a **new representation** of an image
- Can we try to **optimize the parameters** of a convolutional network for image classification?
- First let us see a typical convolutional neural network

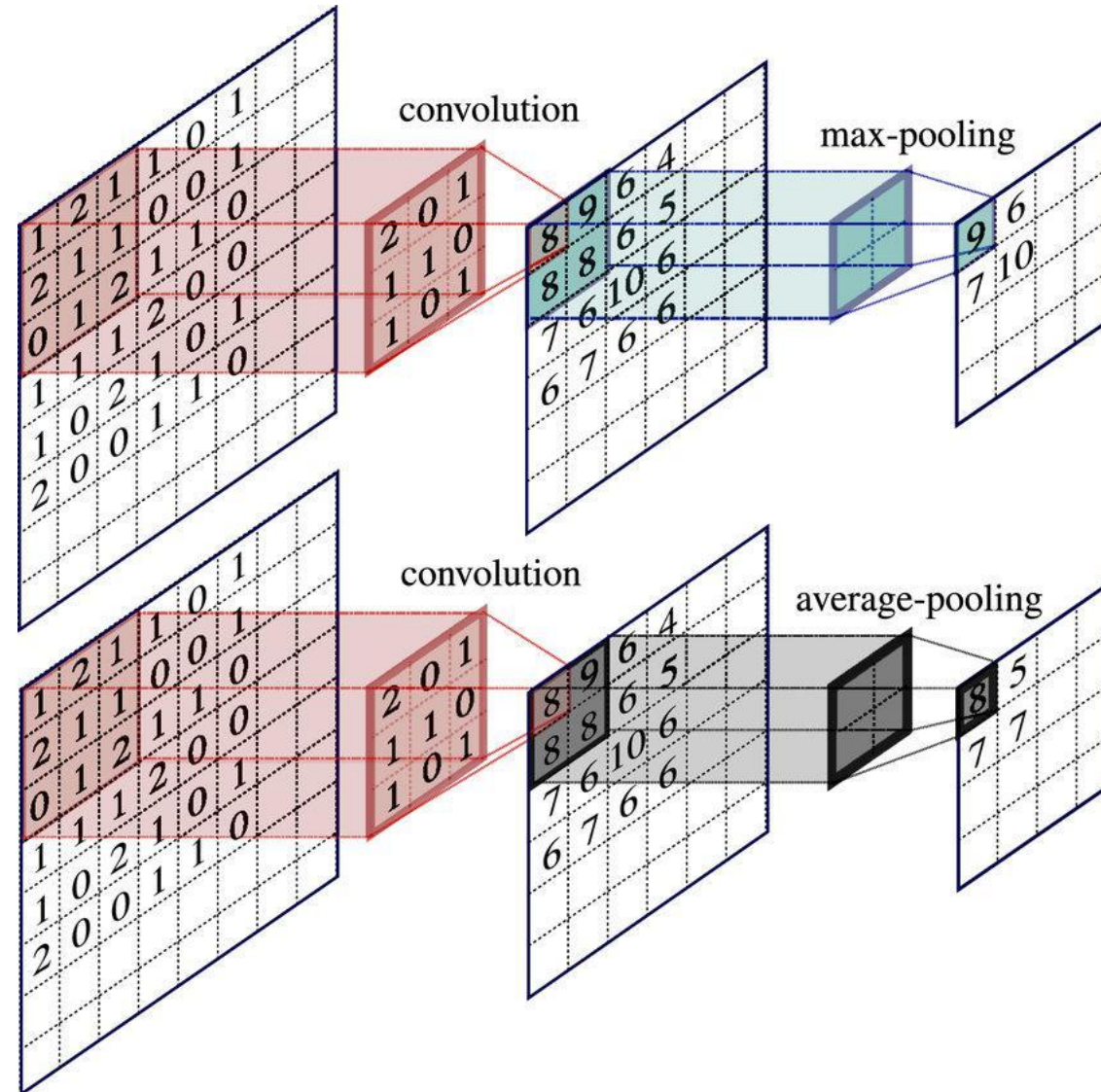
A Typical CNN



Convolutional neural network (CNN)

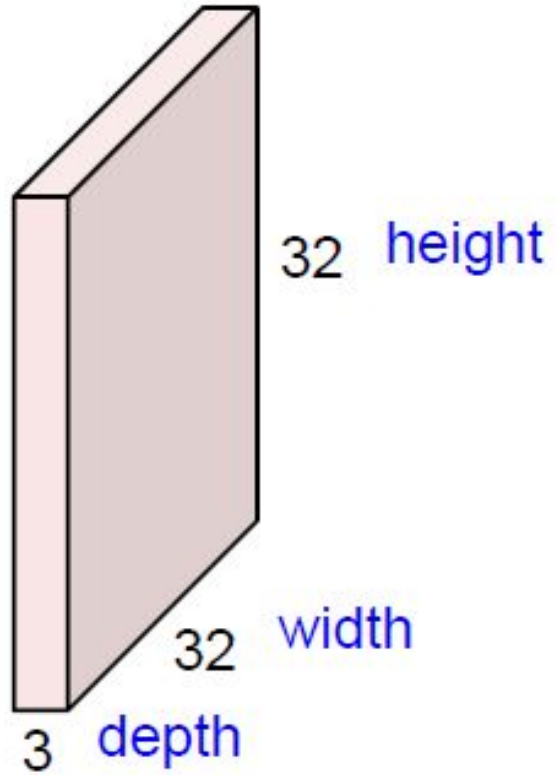
- We see that a CNN has two main components: **convolutions** and **pooling**
- **Convolutions** serve to **partition the image** and look at **local regions** instead of the entire image as given
- **Pooling reduces dimensionality** which helps in the optimization of parameters

Convolution & Pooling



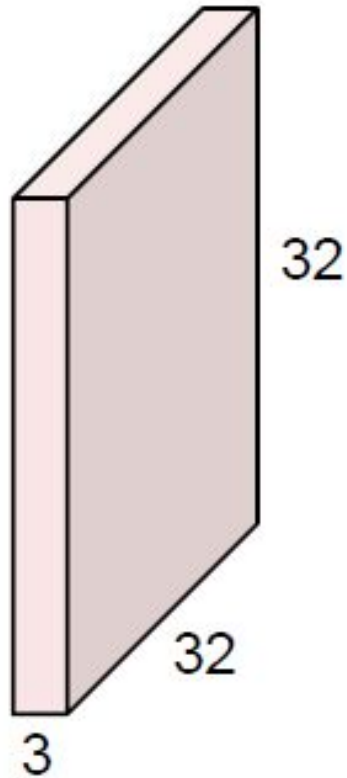
Convolution Layer

32x32x3 image

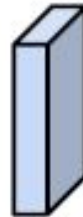


Convolution Layer

32x32x3 image

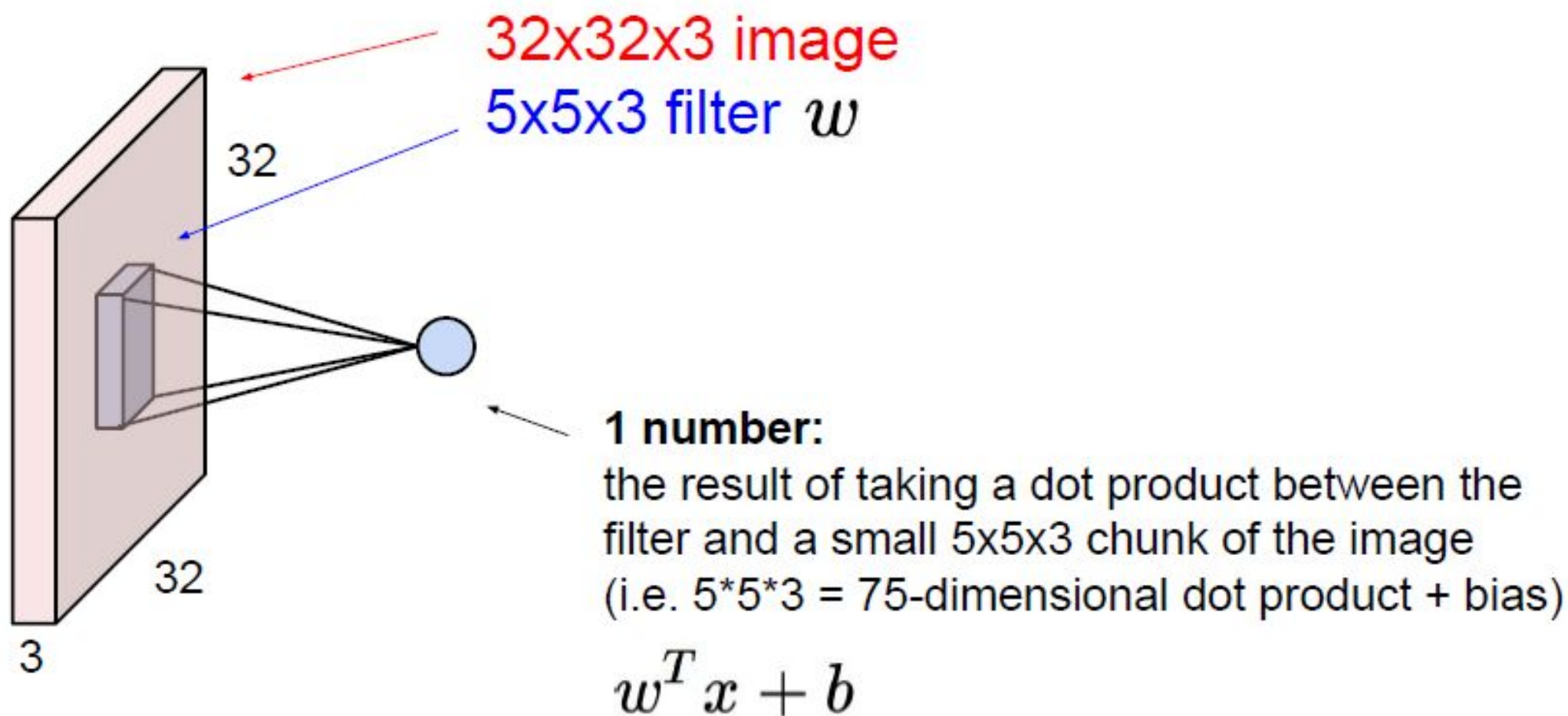


5x5x3 filter

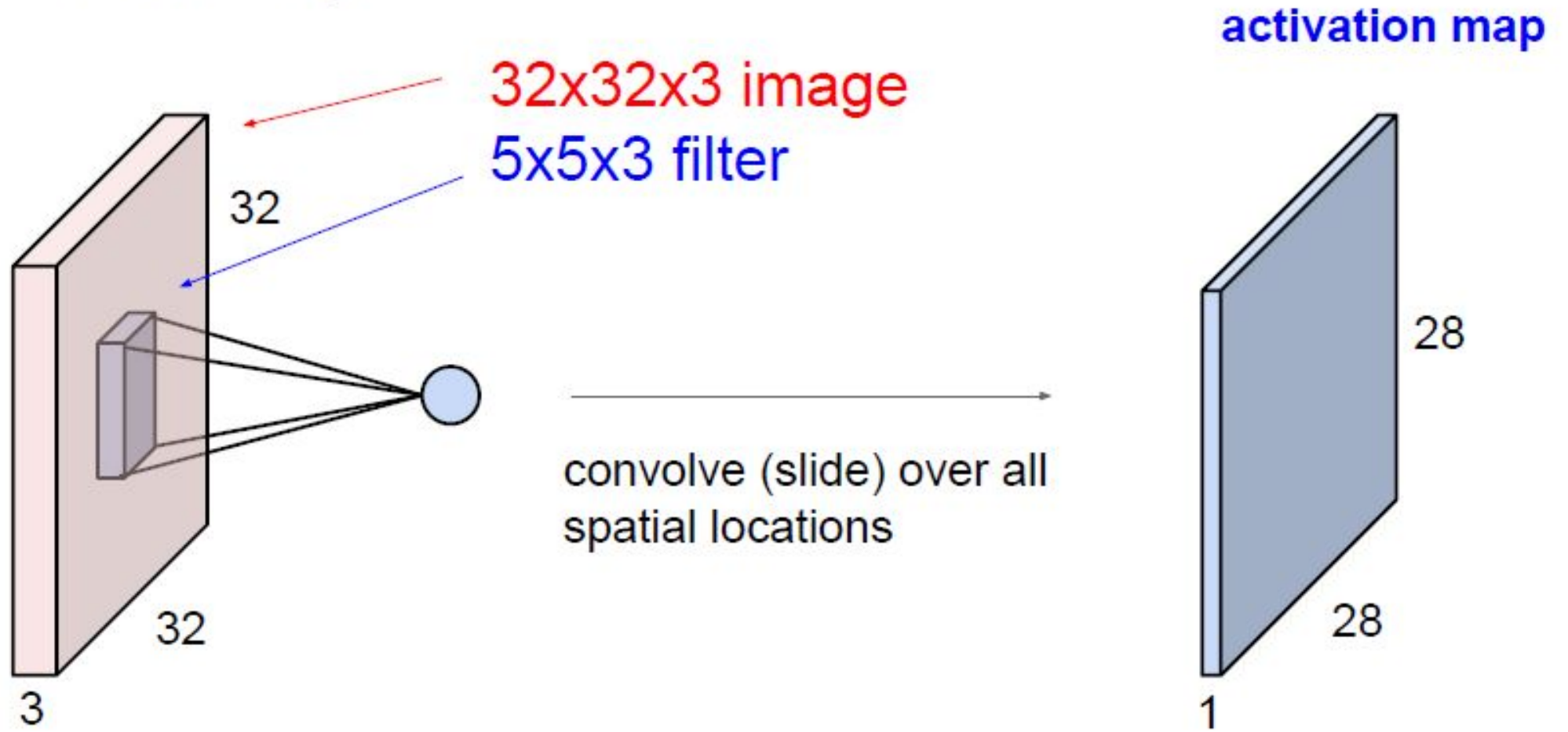


Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

Convolution Layer

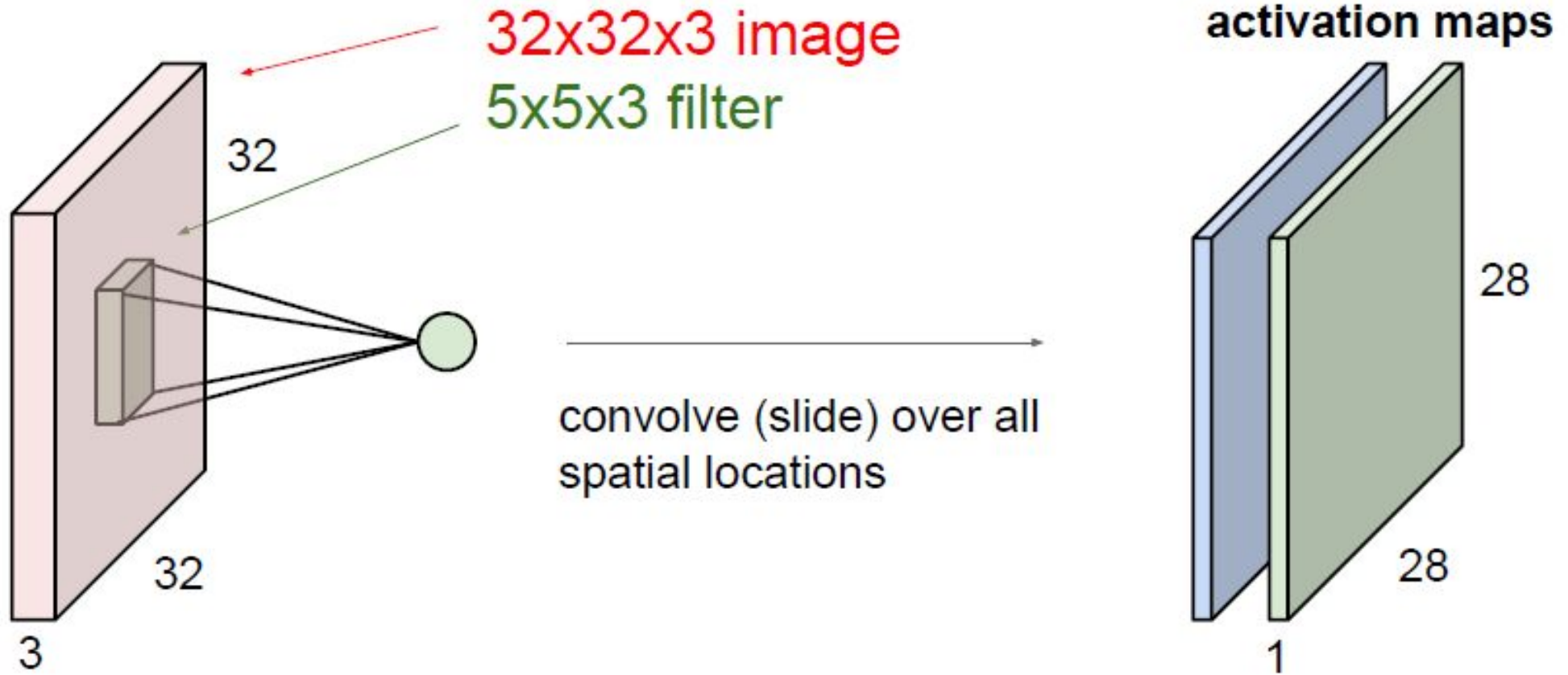


Convolution Layer

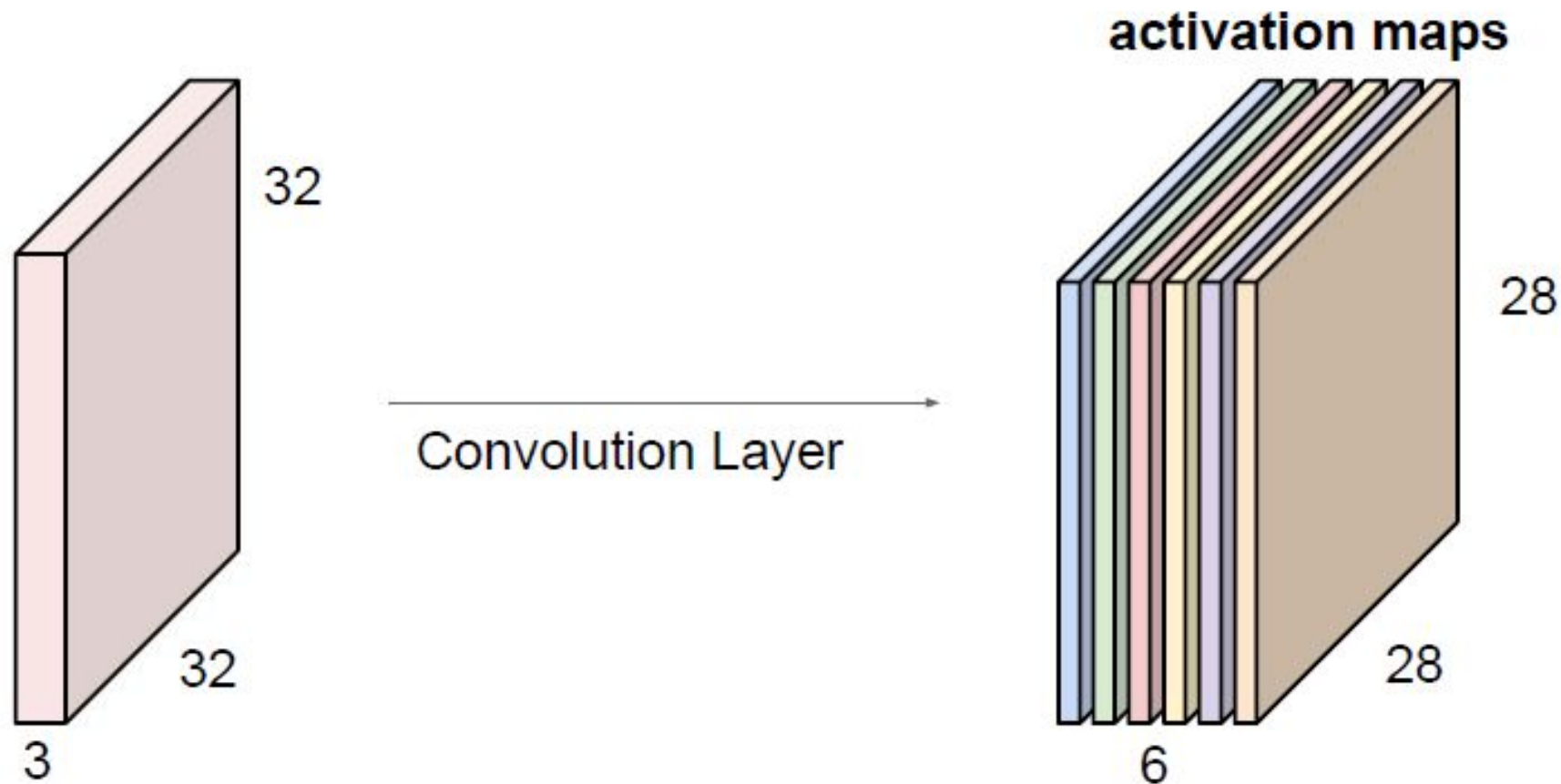


Convolution Layer

consider a second, **green** filter



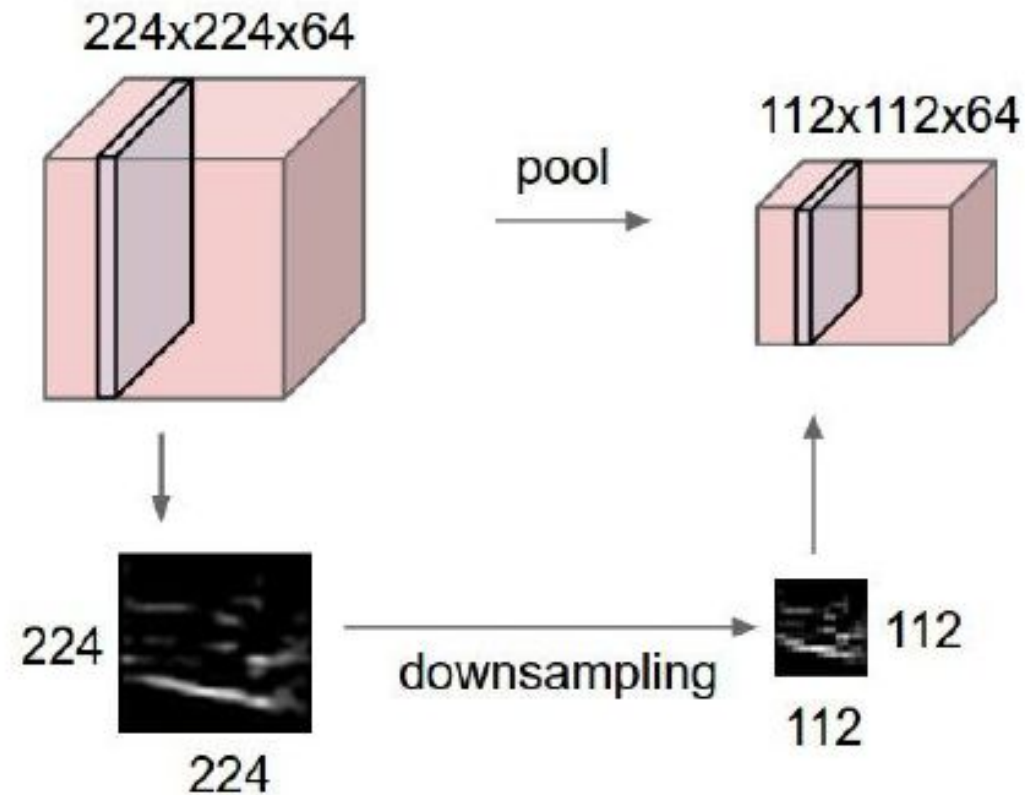
For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



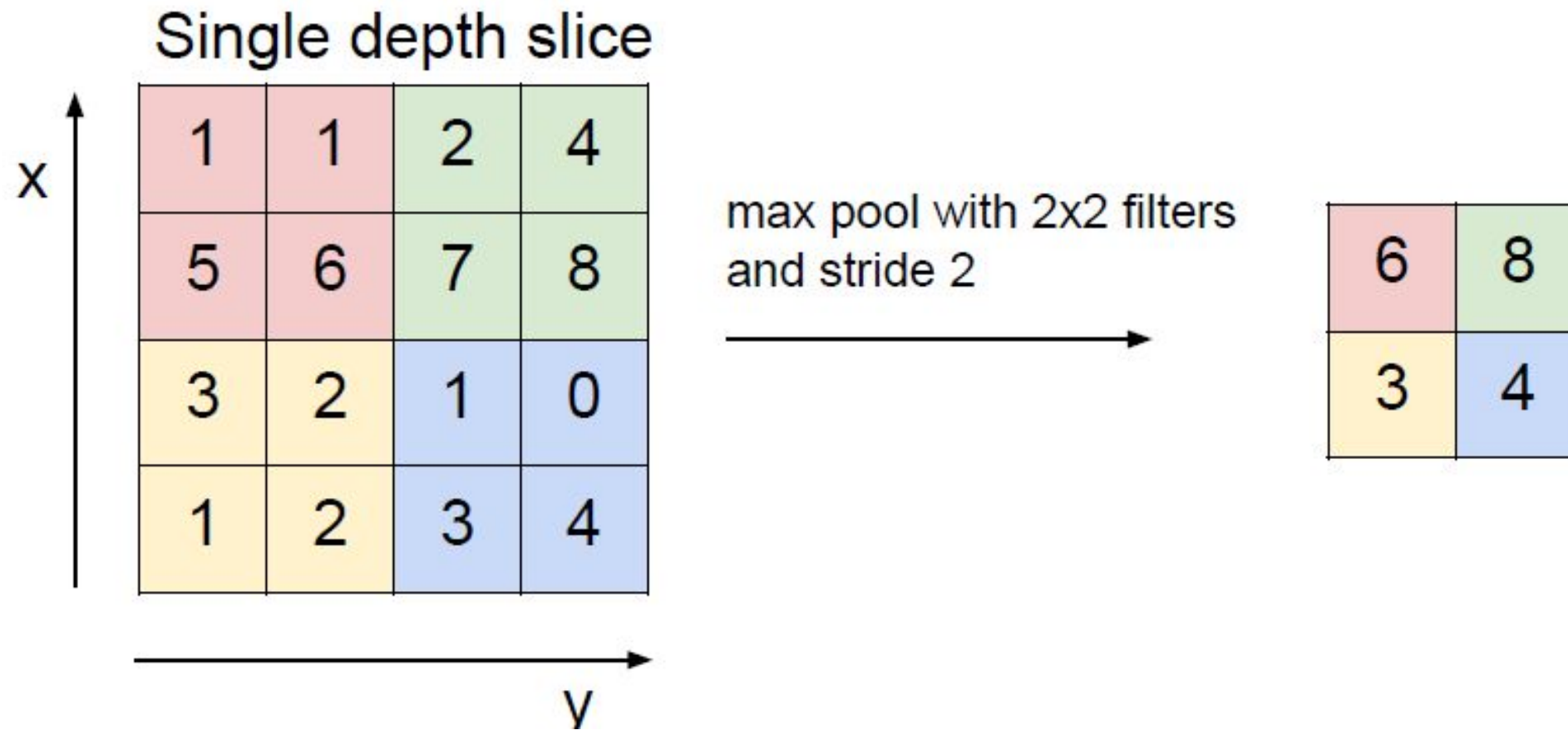
We stack these up to get a “new image” of size 28x28x6!

Pooling layer

- makes the representations smaller and more manageable
- operates over each activation map independently:



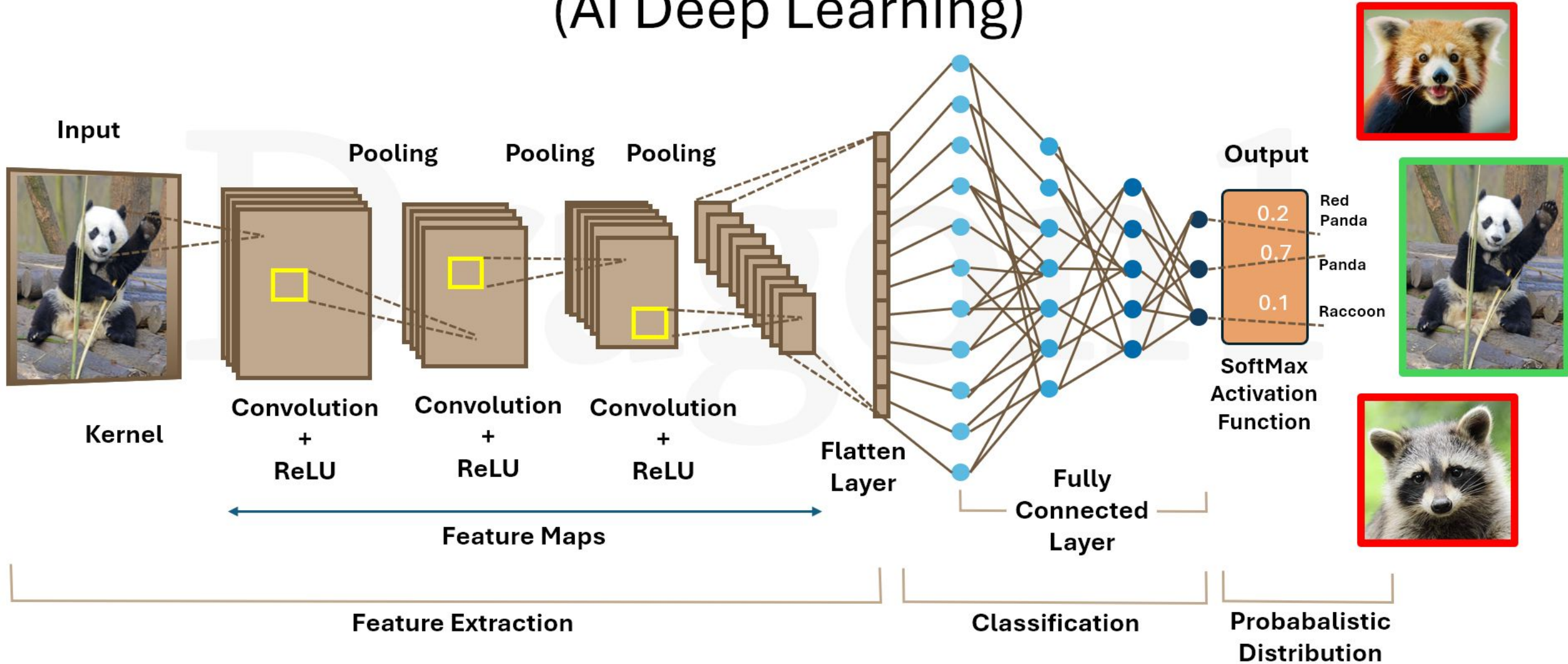
MAX POOLING



Convolutional Neural Networks

(AI Deep Learning)

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Questions?